

The Transpose of a Matrix

Fold

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The Transpose of a Matrix

Definition: If A is an $m \times n$ matrix, then the **Transpose of A** denoted A^T is the $n \times m$ matrix resulting from interchanging both the rows and columns of A , that is $A^T = (a_{ji})$.

For example, suppose that we have the following matrix A . If we interchange the rows and columns of A , then we will obtain the matrix A^T .

One important point to note is that if A is a square matrix, then the diagonal entries of A^T will be the same as that of A . For example, if $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, then $A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$. Notice that the entries in the main diagonal are the same between both matrices.

Example 1

Given the following matrix, determine A^T .

There are 2 rows and 5 columns to A , hence A is a 2×5 matrix, and A^T will be a 5×2 matrix. Interchanging the rows and columns of A we obtain:

(1)

Example 2

Prove that $A + A^T$ is only defined if A is a square matrix.

- **Proof:** Suppose not, and let A be a matrix of size $m \times n$ where $m \neq n$. It thus follows that A^T will be of size $n \times m$. Recall that matrix addition is only defined if all matrices in the sum are the same size. Hence $A + A^T$ is not defined since $m \neq n$.

(2)